

Transition : DS & Algs

Intro
Logic



First: Intro & overview

- Syllabus (updating)
↳ HW vs Quizzes?
- Survey (by Wednesday if possible)
- Class structure: readings + lecture / discussion

Course Goals

Dfn: A proposition is a declarative statement which is either true or false.

Ex:

- This is transition to computing.
- I am a professor.

Non-examples:

- Spiders are creepy.
- Go pick that up.

Note: English can be tricky!

Operations on propositions

Let p be a proposition.

ie: $p = \text{"The sky is blue"}$

- Negation of p : $\neg p$

"It is not the case that p "

or, shorter: "The sky is not blue"

- Conjunction: $p \wedge q \rightarrow p \text{ and } q$

True if both p and q are true.

- Disjunction: $p \vee q \rightarrow p \text{ or } q$

True if either p or q is true.

Truth tables

Summary of truth values of compound propositions for any combination of inputs.

Example:

P	q	$P \vee q$	$P \wedge q$
T	T		
T	F		
F	T		
F	F		

Why? Can help break down complex situations, & can help with logical equivalence ie
 $\neg P \vee \neg q$ vs $P \wedge q$

Other concepts

- Tautology: a proposition that is always true
- Contradiction: a proposition that is always false

Examples:

$$p \vee \neg p$$

$$p \wedge \neg p$$

Table:

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$

• Implications: $p \rightarrow q$ ie $\left\{ \begin{array}{l} \text{"If } p, \text{ then } q\text{"} \\ \text{"} p \text{ implies } q\text{"} \\ \text{"} q \text{ if } p\text{"} \end{array} \right.$

Ex: "If I am elected, then I will lower taxes."

When is this false?

P	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- Consider $p \rightarrow q$:
- Converse: $q \rightarrow p$
 - Inverse: $\neg p \rightarrow \neg q$
 - Contrapositive: $\neg q \rightarrow \neg p$

Example: $p =$ "The number 15 is prime"
 $q =$ "15 has no divisors"

- $p \rightarrow q$: If 15 is prime, then it has no divisors.
- $q \rightarrow p$:
- $\neg p \rightarrow \neg q$:
- $\neg q \rightarrow \neg p$:

Truth table

P	q	$P \rightarrow q$	$q \rightarrow P$	$\neg P$	$\neg q$	$\neg P \rightarrow \neg q$	$\neg q \rightarrow \neg P$
T	T	T	T	F	F	T	T
T	F	F	F	F	T	F	T
F	T	T	T	T	F	T	F
F	F	T	F	T	T	F	T

Which are equivalent?

Using logical equivalence

We can develop some useful "laws" using logical equivalence!

Example: De Morgan's Law

P	Q	$P \vee Q$	$\neg(P \vee Q)$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$
T	T					
T	F					
F	T					
F	F					

Others

Idempotent laws:	$p \vee p \equiv p$	$p \wedge p \equiv p$
Associative laws:	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Commutative laws:	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Distributive laws:	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Identity laws:	$p \vee F \equiv p$	$p \wedge T \equiv p$
Domination laws:	$p \wedge F \equiv F$	$p \vee T \equiv T$
Double negation law:	$\neg\neg p \equiv p$	
Complement laws:	$p \wedge \neg p \equiv F$ $\neg T \equiv F$	$p \vee \neg p \equiv T$ $\neg F \equiv T$
De Morgan's laws:	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
Absorption laws:	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Conditional identities:	$p \rightarrow q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Predicates

propositions which depend on a variable
(or possibly several)

Written $P(x)$ ie $P(x) = "x \text{ is } > 0"$

$Q(x, y) = "x \text{ is a teacher of class } y"$

$R(x) = "x \text{ is prime}"$

Truth value depends on variables:

ie: $P(5) =$

$P(-2) =$

$Q("Erin", "CSE 50502") =$

Quantifiers

- Universal quantifier:

$$\forall x P(x)$$

↳ For all x in domain, $P(x)$ is true

Example:

$$\forall x \in \mathbb{Z}, x \text{ is prime.}$$

Counter example:

To show false, find one element
in domain where $P(x)$ is false

• Existential quantifier

$$\exists x P(x)$$

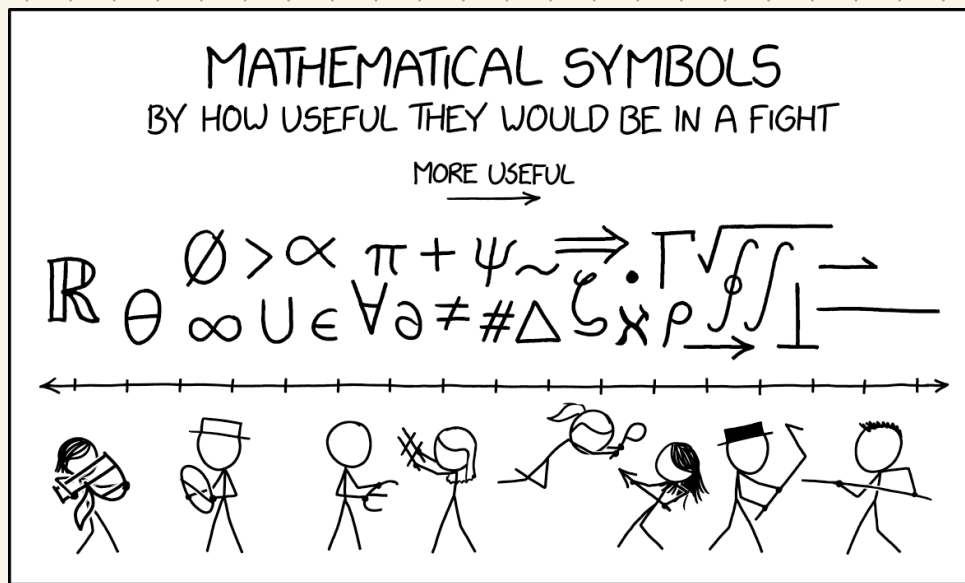
↳ "There exists a value x in the domain,
such that $P(x)$ is true"

Example: $\exists x \in \mathbb{Z}$ st. x is prime.

How to show it's true?

Quick note

At this point, let me pause and acknowledge that math notation can be a lot.



Please stop me with questions!

Using Logical operators

Can combine predicates using all the same logical operators!

$$P(x) = "x + 1 > 2"$$

$$Q(x) = "x < 2"$$

Example:

$$\forall x \in \mathbb{R}, P(x) \vee \neg Q(x).$$

Translate:

Ordering & combining

In short: order matters!

Let $Q(x, y) = "x + y = 0"$

Translate

- $\forall x \exists y$ s.t. $Q(x, y)$

- $\exists x \forall y$ s.t. $Q(x, y)$:

Negations

Consider $P(x) = "x \text{ has taken programming}"$

$\forall \text{ people } x, P(x) :$

What is $\neg(\forall x P(x))$?

Reverse it: What is $\neg(\exists x P(x))$?

They "stack", plus can use laws:

- $\neg(\forall x \forall y P(x, y))$:

- $\neg(\forall x (P(x) \vee Q(x)))$:

Implications & quantifiers

Let $P(x) = "x > 0"$

and $Q(x, y) = "xy \leq 0"$

translate:

$$\forall x \forall y (P(x) \wedge \neg P(y) \rightarrow Q(x, y))$$

Proofs

A theorem (or lemma, proposition, etc.) is a statement that can rigorously be shown to hold true.

Generally in math, something like:

"If n is even, then it is divisible by 2."

" $\forall \varepsilon > 0 \exists \delta > 0$ s.t. if $|f(x) - f(y)| < \varepsilon$,
then $|x - y| < \delta$ "

The sequence of statements giving the argument is a proof.